

Problema 1:

$$W_{AB} = \int_{V_A}^{V_B} p dV = nRT_A \ln\left(\frac{V_B}{V_A}\right) = p_A V_A \ln\left(\frac{V_B}{V_A}\right)$$

$$V_B = V_A e^{\frac{W_{AB}}{p_A V_A}} = 0,0156 \text{ m}^3$$

$$Q_{CA} = n C_V (T_A - T_C); T_B V_B^{\frac{C_P}{C_V}-1} = T_C V_C^{\frac{C_P}{C_V}-1}; C_V = \frac{5}{2} R$$

$$Q_{CA} = \frac{5}{2} p_A V_A \left(1 - \left(\frac{V_B}{V_A} \right)^{\frac{C_P}{C_V}-1} \right) = -4377,6 \text{ J}$$

Problema 2:

$$\eta = 0,9 \left(1 - \frac{T_2}{T_1} \right) = \frac{W}{Q_1} = \frac{W}{W + |Q_2|} \Rightarrow |Q_2| = W \left(\frac{T_1}{0,9(T_1 - T_2)} - 1 \right)$$

$$|Q_2| = 3407 \text{ J}$$

Problema 3: Si llamamos Φ_1 al flujo del campo eléctrico a través de la superficie de área A y Φ_2 al flujo a través del resto de la superficie cilíndrica, por la Ley de Gauss debe ser:

$$\Phi_1 + \Phi_2 = \frac{q}{\epsilon_0} \Rightarrow \Phi_2 = \frac{q}{\epsilon_0} - \Phi_1 = \left(\frac{8,85 \times 10^{-9}}{8,85 \times 10^{-12}} - 100 \right) \frac{\text{N} \cdot \text{m}^2}{\text{C}} = 900 \frac{\text{N} \cdot \text{m}^2}{\text{C}}$$

Problema 4:

$$a) V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{d_1} + \frac{q}{d_2} \right) \quad d_1 = 3 \text{ m} \text{ y } d_2 = 5 \text{ m}$$

$$q = 4\pi\epsilon_0 V \left(\frac{d_1 d_2}{d_1 + d_2} \right) \quad q = 1,88 \cdot 10^{-7} \text{ C}$$

b) Debe justificar que el campo es nulo en $(2;0) \text{ m}$

Problema 5:

A partir de la ley de Ampère debe justificar que en un solenoide ideal:

$$B = \mu_0 n I \Rightarrow n = \frac{B}{\mu_0 I} \Rightarrow n = 1592 \text{ espiras / m}$$

Problema 6:

$$ei = -\frac{d\phi}{dt} \quad \phi = -B\pi R_{(t)}^2 \quad ei = -\frac{d(-B\pi R_{(t)}^2)}{dt} = B\pi \frac{dR_{(t)}^2}{dt}$$

$$ei = B\pi R \frac{dR}{dt} \Rightarrow \frac{ei}{B\pi} dt = R dR \Rightarrow \int_0^t \frac{ei}{B\pi} dt = \int_{R_0}^{R(t)} R dR$$

$$\frac{ei}{B\pi} t = \frac{1}{2} (R^2 - R_0^2) \Rightarrow R = \sqrt{\frac{2ei}{B\pi} t + R_0^2}$$

$$R = \sqrt{4t + 0,25} \text{ [m;s]}$$

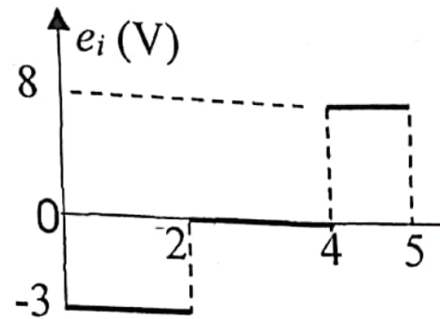
Problema 7:

$$e_i = -\frac{d\phi}{dt} = -\frac{d(L I_{(t)})}{dt} = -L \frac{d I_{(t)}}{dt}$$

$$0 < t < t_1 \quad e_i = -2 \frac{4-1}{2-0} \text{ V} = -3 \text{ V}$$

$$t_1 < t < t_2 \quad e_i = 0$$

$$t_2 < t < t_3 \quad e_i = -2 \frac{0-4}{5-4} \text{ V} = 8 \text{ V}$$

**Problema 8:**

$$X_L = 800 \, \Omega, \varphi = -\frac{\pi}{3}, I_{ef} = 2 \text{ A}, L = 0,6 \text{ H}, f = 150 \text{ Hz}$$

$$a) \omega = 2\pi f = 300\pi \frac{1}{\text{s}}$$

$$X_L = \omega \cdot L = 565,48 \, \Omega \Rightarrow X = X_L - X_C = -234,51 \, \Omega$$

$$\text{tg}(\varphi) = \frac{X}{R} \Rightarrow R = \frac{X}{\text{tg}(\varphi)} = \frac{-234,51 \, \Omega}{\text{tg}(-\frac{\pi}{3})} = 135,39 \, \Omega$$

$$\underline{\underline{R = 135,39 \, \Omega}}$$

$$b) V_{ef} = Z \cdot I_{ef} = \sqrt{R^2 + X^2} I_{ef}$$

$$Z = 270,79 \, \Omega$$

$$\underline{\underline{V_{ef} = 541,58 \text{ V}}}$$